

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of BSc (Biological Sciences) Examination March 2019

Course code & Name: BS 304 Fundamentals of Microbiology II

Date: 14/03/2019 Day: Thursday Time: 1:30 PM To 1:50 PM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q - I Choose the correct answer for the following questions.

10

1. The cell division by binary fission begins with the following
(i) mesosome attachment (ii) equatorial ridge formation
(iii) septum formation (iv) all of the above
2. Bacterial cell bursts in the following medium
(i) hypotonic (ii) acidic
(iii) hypertonic (iv) alkaline
3. The following indicates that all the cells present in a culture are in the same stage of its life cycle.
(i) Continuous (ii) Fed batch
(iii) Batch (iv) synchronous
4. The following test is most suitable for determination of the sensitivity of an organism to an antibiotic
(i) phenol coefficient (ii) E test
(iii) disc diffusion (iv) all of the above

5. The following disinfectant can kill microorganisms by the generation of superoxide radicals
- | | |
|----------------------|-------------------|
| (i) ethanol | (ii) formaldehyde |
| (iii) gluteraldehyde | (iv) peroxygens |
6. Rubber gloves may be sterilized by _____.
- | | |
|-----------------|-------------------|
| (i) autoclaving | (ii) filtration |
| (iii) radiation | (iv) hot air oven |
7. The following group of microorganisms can grow between 50°C to 80°C.
- | | |
|-------------------------|-----------------------|
| (i) thermophiles | (ii) mesophiles |
| (iii) hyperthermophiles | (iv) all of the above |
8. The following is (are) the direct method(s) for determination of microbial growth
- | | |
|------------------|-----------------------|
| (i) turbidity | (ii) plate count |
| (iii) dry weight | (iv) all of the above |
9. The microflora that establish themselves briefly, but are excluded by competition from normal flora or by human immune system is called
- | | |
|----------------|-----------------------|
| (i) permanent | (ii) transient |
| (iii) resident | (iv) all of the above |
10. Biocontrol agent may act by following the number of pests in the field
- | | |
|------------------|-----------------------|
| (i) increasing | (ii) equalizing |
| (iii) decreasing | (iv) all of the above |

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III Semester of BSc (Biological Sciences) Examination March 2019

Course code & Name: BS 304 Fundamentals of Microbiology II

Date: 14/03/2019 Day: Thursday Time: 1:50 P M To 3:30 P M Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

- Q - II Answer the following questions as directed 10**
- A Discuss any two of the following (06)**
- i. Microbial loop with a labeled diagram
 - ii. Normal flora of human body
 - iii. Types of Biofertilizer
- B Attempt any two of the following: (04)**
- i. Write a short note on: Probiotics
 - ii. List the methods used for sampling of air. Discuss any one in brief.
 - iii. Discuss briefly the microorganisms present in soil

SECTION - II

- Q-III Answer the following questions as directed 15**
- A Define any two the following terms (02)**
- i. Microbial growth
 - ii. Generation time
 - iii. Turbidostat
- B Attempt any two the following (04)**
- i. Classify microorganisms on the basis of oxygen requirement
 - ii. List the phases of bacterial growth and discuss log phase in brief.
 - iii. Discuss briefly the growth of microorganism in continuous culture
- C List the indirect methods of measurement of growth (01)**
- D Differentiate between any two the following (02)**
- i. Autoclaving and hot air oven as a method of sterilization
 - ii. Bactericidal and bacteriostatic agents
 - iii. Batch culture and Fed batch culture
- E Attempt any two of the following (06)**
- i. Explain tyndelization in brief
 - ii. Write a short note on: Use of Peroxygens as disinfectant
 - iii. List the methods of evaluating antibiotics. Discuss one in detail